

A Review of Eptinezumab Use in Migraine.

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We present a review of the efficacy and safety profile of eptinezumab (also known by the brand name Vyepti), a calcitonin gene-related peptide (CGRP) monoclonal antibody (mAb) developed by Lundbeck Seattle BioPharmaceuticals, Inc., that received its first approval in the USA on 21 February 2020 for the preventive treatment of migraine in adults. It is administered by an intravenous infusion at a 100 mg dose every 3 months and shows no drug interactions. Studies have shown that eptinezumab is an effective preventative medication in migraine which starts showing its effect from day 1 of its administration, which maintains a consistent level of efficacy through a year of its treatment at doses 100 mg and 300 mg. It was found to be effective at reducing time to headache pain freedom during acute migraine attacks as well. Eptinezumab is a relatively safe drug for the prevention of migraines with treatment-related adverse events occurring at a low frequency. They bear a safe profile in patients with comorbidities like obesity and type 1 diabetes. The most frequent adverse events observed were nasopharyngitis, upper respiratory tract infections (URTIs), and sinusitis and were usually mild. The development of anti-drug antibodies was common, but they declined to undetectable levels with continued dosing and did not appear to impact the overall safety profile of the drug. Further studies are needed to assess long-term safety, use in different patient populations, and to compare its efficacy to other drugs of its class.

Safety and tolerability of monoclonal antibodies targeting the CGRP pathway and gepants in migraine prevention: A systematic review and network meta-analysis.

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Direct comparisons of the tolerability and safety of migraine preventive treatments targeting the calcitonin gene-related peptide pathway are lacking. This study aimed to compare the safety and tolerability of anti-calcitonin gene-related peptide monoclonal antibodies and gepants in migraine prevention. A network meta-analysis of phase 3 randomized controlled trials assessing the safety and tolerability of anti-calcitonin gene-related peptide monoclonal antibodies (erenumab, eptinezumab, fremanezumab, or galcanezumab) and gepants (atogepant, rimegepant) in migraine prevention was performed. Primary outcomes were treatment-emergent adverse events and serious adverse events. Secondary outcomes included any adverse events, adverse events leading to treatment discontinuation and individual adverse events. We included 19 randomized controlled trials, comprising 14,584 patients. Atogepant 120 mg (OR 2.22, 95% CI [1.26, 3.91]) and galcanezumab 240 mg (OR 1.63, 95% CI [1.33, 2.00]) showed the largest odds of treatment-emergent adverse events compared to placebo. While eptinezumab 30 mg had greater odds of adverse events leading to treatment discontinuation (OR 2.62, 95% CI [1.03, 6.66]). No significant differences in serious adverse events were found between active treatments and placebo. Eptinezumab was associated with the lowest odds of treatment-emergent adverse events and serious adverse events compared to placebo, whereas erenumab was associated with the lowest odds of any adverse events and quarterly fremanezumab with the lowest odds of treatment discontinuation due to adverse events. Monoclonal antibodies targeting the calcitonin gene-related peptide pathway and gepants are a safe and well tolerated option for migraine prevention.

Safety and efficacy of eptinezumab for migraine prevention in patients with two-to-four previous preventive treatment failures (DELIVER): a multi-arm, randomised, double-blind, placebo-controlled, phase 3b trial.

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The monoclonal antibody eptinezumab, which targets calcitonin gene-related peptide, has shown migraine preventive effects starting the day following infusion and acceptable safety and tolerability in phase 3 trials, but benefits in the subpopulations of patients with previous preventive treatment failures were not examined. We aimed to investigate the safety and efficacy of eptinezumab for migraine prevention in adults with migraine and two-to-four previous preventive treatment failures. DELIVER was a multicentre, multi-arm, phase 3b trial comprising a 24-week double-blind, placebo-controlled period and a 48-week dose-blinded extension. We recruited adults with episodic or chronic migraine with at least 4 monthly migraine days (as per International Headache Society guidelines) and documented evidence of two-to-four previous preventive treatment failures within the past 10 years, from 96 study locations across Europe (n=93) and the USA (n=3). Patients were randomly assigned (1:1:1) via a centralised randomisation system, stratified by baseline monthly headache days and country, to eptinezumab 100 mg, eptinezumab 300 mg, or placebo. The primary efficacy endpoint was the change from baseline in mean monthly migraine days (captured using a daily electronic diary) in weeks 1-12, assessed in the full analysis set. All participants and study personnel were masked to study drug assignments. The dose-blinded extension period is ongoing. The trial is registered with ClinicalTrials.gov, NCT04418765, and EudraCT, 2019-004497-25. Between June 1, 2020, and Oct 7, 2021, 891 individuals were randomly assigned and received at least one dose of study drug (safety population; eptinezumab 100 mg n=299 [34%], eptinezumab 300 mg n=294 [33%], placebo n=298 [33%]). 865 patients completed the placebo-controlled period. The change from baseline to weeks 1-12 in mean monthly migraine days was -4.8 (SE 0.37) with eptinezumab 100 mg, -5.3 (0.37) with eptinezumab 300 mg, and -2.1 (0.38) with placebo. The difference from placebo in change in mean monthly migraine days from baseline was significant with eptinezumab 100 mg (-2.7 [95% CI -3.4 to -2.0]; $p<0.0001$) and eptinezumab 300 mg (-3.2 [-3.9 to -2.5]; $p<0.0001$). Treatment-emergent adverse events occurred in 127 (42%) of 299 patients in the eptinezumab 100 mg group, in 120 (41%) of 294 in the eptinezumab 300 mg group, and in 119 (40%) of 298 in the placebo group. The most common treatment-emergent adverse event was COVID-19 (20 [7%] of 299 patients in the eptinezumab 100 mg group, 17 [6%] of 294 in the eptinezumab 300 mg group, and 16 [5%] of 298 in the placebo group). Serious adverse events were uncommon (five [2%] of 299 in the eptinezumab 100 mg group, seven [2%] of 294 in the eptinezumab 300 mg group, four [1%] of 298 in the placebo group) and included anaphylactic reaction (eptinezumab 300 mg n=2) and COVID-19 (eptinezumab 100 mg n=1 and eptinezumab 300 mg n=1). In adults with migraine and two-to-four previous preventive treatment failures, eptinezumab provided significant migraine preventive effects compared with placebo, with acceptable safety and tolerability, indicating that eptinezumab might be an effective treatment option for this patient population. The dose-blinded extension period will provide additional long-term safety data in patients with migraine and previous preventive treatment failures. H Lundbeck.

Calcitonin-gene-related peptide pathway mAbs and migraine prevention.

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Monoclonal antibodies (mAbs) targeting the calcitonin-gene-related peptide (CGRP) pathway have been

developed for episodic and chronic migraine prevention, either through binding the CGRP ligand (eptinezumab, fremanezumab, galcanezumab) or the CGRP receptor (erenumab). We provide an update on published Phase 2 and Phase 3 trials, safety/tolerability data, pharmacokinetics and mechanism of action of these biologicals. The efficacy data from Phase 2 trials are corroborated by those from published Phase 3 trials, with a multitude of publications expected in 2018. Review of safety data concluded there was no difference in total adverse events or main adverse events (including upper respiratory tract infection, nasopharyngitis, nausea, injection-site pain and back pain) between the mAbs and placebo injections except apparently for dizziness. The site of action of these mAbs is not fully elucidated but current insight is that their effect resides in the periphery; a contribution of central effect(s) can however not be excluded at present. Although efficacy of all four drugs is modest over placebo in episodic and chronic migraine prevention and overall comparable with available oral preventive treatments, current tolerability and (short-term) safety data of this new treatment approach certainly promise a major step forward for migraine patients.

Advances in CGRP Monoclonal Antibodies as Migraine Therapy: A Narrative Review.

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Migraine is a potentially disabling disorder, yet it remains underdiagnosed and undertreated. The release of the neuropeptide calcitonin gene-related peptide (CGRP) in the trigemino-cerebrovascular system plays a vital role in the evolution of migraine. It enhances peripheral sensitization by mediating neurogenic inflammation and also influences central sensitization. The majority of the drug classes available for migraine prophylaxis are nonspecific and associated with numerous side effects and drug interactions. Anti-CGRP monoclonal antibodies (mAb) are an innovative therapeutic class that fulfills the need for more efficacious and tolerable preventive therapy. While erenumab is a mAb to the CGRP receptor, eptinezumab, fremanezumab, and galcanezumab bind to the CGRP molecule. They decrease the number of headache days and improve disability. Upper respiratory tract infection, nausea, constipation, pain at the site of injection, and fatigue are the associated side effects. CGRP mAbs are an excellent advancement in translational research and are a promising addition in migraine therapy. This article discusses the recent advances in the development of the CGRP mAbs.

Prevention of migraine with monoclonal antibodies against CGRP or the CGRP receptor: Addition to the S1 guideline: Therapy of migraine attacks and prevention of migraine. Recommendations of the Germany Society of Neurology and the German Migraine and Headache Society.

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Monoclonal antibodies against the calcitonin gene-related peptide (CGRP) receptor (Erenumab) or against CGRP (Eptinezumab, Fremanezumab, Galcanezumab) are new substances for the preventive treatment of migraine. They represent an extension of the therapeutic options, which already exist in migraine prevention. In randomized, placebo-controlled studies, the efficacy and good tolerability of these specific substances have been demonstrated in patients with episodic and chronic migraine. The following treatment recommendation presents a summary of the pivotal studies. Recommendations are provided for the targeted selection of patients as well as for the evaluation of therapeutic success and the duration

of treatment. Finally, possible restrictions on the use of this new substance group are discussed. This guideline is an abridged and translated version of the guideline published by Diener H-C, May A et al., Prevention of migraine with monoclonal antibodies against CGRP or the CGRP receptor, Supplement to S1 Guideline Therapy of Migraine Attack and Prevention of Migraine, 2019, Deutsche Gesellschaft für Neurologie (eds.), Guidelines for Diagnostics and Therapy in Neurology. A complete version of this guideline can be found on the website of the Deutsche Gesellschaft für Neurologie (www.dgn.org/leitlinien) and the AWMF (Arbeitsgemeinschaft wissenschaftlicher Medizinischer Gesellschaften). This guideline has been approved by the German Neurological Society (DGN) and the German Migraine and Headache Society (GMHS) and was reviewed by the two societies.

Could the New Anti-CGRP Monoclonal Antibodies Be Effective in Migraine Aura? Case Reports and Literature Review.

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Recently, monoclonal antibodies (mAbs) directed against calcitonin gene-related peptide (CGRP) (Eptinezumab, Fremanezumab, and Galcanezumab) or its receptor (Erenumab) have been approved for clinical use as prophylactic drugs for high-frequency episodic and chronic migraine. While their therapeutic effects on headache pain is well documented, there is scarce information on the usefulness of these medications in preventing migraine aura, which is believed to be associated with cortical spreading depression (CSD). Because of their large size, mAbs cannot easily cross the blood-brain barrier in high quantities, rendering the peripheral trigeminovascular system to likely be a major site of their action. In this paper, we report two cases of patients suffering from migraine with and without aura, who reported a complete disappearance of aura or reduced aura duration and intensity while taking Galcanezumab or Erenumab, respectively. Then, we present a brief overview of the literature about the controversial relationship between CSD and CGRP and about the potential "additional central" role of these mAbs in the pathophysiology of migraine aura.

European headache federation guideline on the use of monoclonal antibodies acting on the calcitonin gene related peptide or its receptor for migraine prevention.

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Monoclonal antibodies acting on the calcitonin gene-related peptide or on its receptor are new drugs to prevent migraine. Four monoclonal antibodies have been developed: one targeting the calcitonin gene-related peptide receptor (erenumab) and three targeting the calcitonin gene-related peptide (eptinezumab, fremanezumab, and galcanezumab). The aim of this document by the European Headache Federation (EHF) is to provide an evidence-based and expert-based guideline on the use of the monoclonal antibodies acting on the calcitonin gene-related peptide for migraine prevention. The guideline was developed following the Grading of Recommendation, Assessment, Development, and Evaluation (GRADE) approach. The working group identified relevant questions, performed systematic review and analysis of the literature, assessed the quality of available evidence, and wrote recommendations. Where the GRADE approach was not applicable, expert opinion was provided. We found low to high quality of evidence to recommend eptinezumab, erenumab, fremanezumab, and galcanezumab in patients with episodic migraine and medium to high quality of evidence to recommend

erenumab, fremanezumab, and galcanezumab in patients with chronic migraine. For several clinical questions, there was not enough evidence to provide recommendations using the GRADE approach and recommendations relied on experts' opinion. Monoclonal antibodies acting on the calcitonin gene-related peptide are new drugs which can be recommended for migraine prevention. Real life data will be useful to improve the use of those drugs in clinical practice.

CGRP as the target of new migraine therapies - successful translation from bench to clinic.

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Treatment of migraine is on the cusp of a new era with the development of drugs that target the trigeminal sensory neuropeptide calcitonin gene-related peptide (CGRP) or its receptor. Several of these drugs are expected to receive approval for use in migraine headache in 2018 and 2019. CGRP-related therapies offer considerable improvements over existing drugs as they are the first to be designed specifically to act on the trigeminal pain system, they are more specific and they seem to have few or no adverse effects. CGRP receptor antagonists such as ubrogepant are effective for acute relief of migraine headache, whereas monoclonal antibodies against CGRP (eptinezumab, fremanezumab and galcanezumab) or the CGRP receptor (erenumab) effectively prevent migraine attacks. As these drugs come into clinical use, we provide an overview of knowledge that has led to successful development of these drugs. We describe the biology of CGRP signalling, summarize key clinical evidence for the role of CGRP in migraine headache, including the efficacy of CGRP-targeted treatment, and synthesize what is known about the role of CGRP in the trigeminovascular system. Finally, we consider how the latest findings provide new insight into the central role of the trigeminal ganglion in the pathophysiology of migraine.

Role of CGRP in Migraine.

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Migraine is a common neurological disorder that afflicts up to 15% of the adult population in most countries, with predominance in females. It is characterized by episodic, often disabling headache, photophobia and phonophobia, autonomic symptoms (nausea and vomiting), and in a subgroup an aura in the beginning of the attack. Although still debated, many researchers consider migraine to be a disorder in which CNS dysfunction plays a pivotal role while various parts of the trigeminal system are necessary for the expression of associated symptoms. Treatment of migraine has in recent years seen the development of drugs that target the trigeminal sensory neuropeptide calcitonin gene-related peptide (CGRP) or its receptor. Several of these drugs are now approved for use in frequent episodic and in chronic migraine. CGRP-related therapies offer considerable improvements over existing drugs, as they are the first to be designed specifically to act on the trigeminal pain system: they are more specific and have little or no adverse effects. Small molecule CGRP receptor antagonists, gepants, are effective for acute relief of migraine headache, whereas monoclonal antibodies against CGRP (Eptinezumab, Fremanezumab, and Galcanezumab) or the CGRP receptor (Erenumab) effectively prevent migraine attacks. The neurobiology of CGRP signaling is briefly summarized together with key clinical evidence for the role of CGRP in migraine headache, including the efficacy of CGRP-targeted treatments.

